

Kavindu Munugoda (PhD)

Research Associate & Natural Product Scientist

Ph.D. qualified Research Associate and Molecular Virologist currently engaged at the Medical Research Institute (MRI), Sri Lanka. Offers over 6 years of cumulative experience in tertiary education and high-impact biomedical research. Proven expertise in bridging traditional ethnobotanical knowledge with modern viral therapeutic development, specifically targeting HPV and vector-borne diseases. Validated history of **delivering undergraduate curriculum**, **supervising Honours research students**, and **securing competitive international funding** (ICGEB). Seeking a University Lecturer or similar position to leverage dual strengths in advanced molecular research and academic teaching.

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Key Competencies

🎓 Tertiary Teaching

Extensive experience delivering guest lectures and tutorials in Research Methodology, Microbiology, and Phytochemistry.

👨‍🔬 Research Supervision

Supervised 6+ undergraduate Honours students through experimental design and thesis submission.

📖 Curriculum Development

Curriculum Development: Designing assessment materials, reading lists, and practical laboratory modules.

🌐 Professional Communication

Strong cross-cultural communication skills honed through international fellowships (China) and industry collaborations (UNDP/WWF).

🔬 Technical Expertise

Possesses robust technical expertise in molecular virology and drug discovery, utilizing mammalian cell culture (HEK293T, HeLa, CaSki, SiHa), cytotoxicity profiling (CCK-8), and dual-luciferase reporter assays to validate therapeutic targets. Proficient in structure-based drug design, including molecular docking (AutoDock Vina, PyRx, UCSF ChimeraX) and protein-ligand interaction analysis. Demonstrated skill in molecular biology techniques such as recombinant DNA technology, molecular cloning, PCR, Western Blotting, and GST-affinity chromatography. Complements these biological skills with strong analytical chemistry capabilities, specifically GC-MS analysis, phytochemical screening, and bioactive compound isolation. Applies advanced data science methodologies to research outcomes, employing Python, R, SPSS, and GraphPad Prism for statistical analysis, multivariate analysis (PCA), and dose-response modeling.

Academic & Research Employment

Department of Vaccine QC and Molecular Biology Research Associate	Medical Research Institute (MRI) – Ministry of Health, Sri Lanka Jan 2026 – Present
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- **Instruction & Training:** Conduct comprehensive training sessions for junior staff on advanced molecular biology techniques, including dual-luciferase reporter assays and cell culture methodologies.
- **Assessment Design:** Evaluate technical competency assessments and laboratory practical examinations for training programs to ensure staff meet research standards.
- **Student Mentorship:** Mentor visiting students and junior researchers on experimental design, troubleshooting, and career development in molecular virology.
- **Research Leadership:** Supervise comprehensive research workflows for junior staff, coordinating experimental planning and milestone tracking.

Molecular Medicine Unit Graduate Research Assistant	University of Kelaniya, Faculty of Medicine, Sri Lanka Dec 2023 – Nov 2024
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- **Teaching Support:** Supported teaching-related laboratory activities, including demonstrations and technical guidance for undergraduate and postgraduate students.
- **Protocol Development:** Contributed to the development, standardization, and optimization of laboratory protocols and Standard Operating Procedures (SOPs) used in teaching.
- **Mentorship:** Supervised and mentored junior research staff and students in laboratory techniques, biosafety practices, and data documentation.

Dean's Office (Applied Sciences)	University of Sri Jayewardenepura, Sri Lanka
Temporary Demonstrator	Sep 2023 – Dec 2023
Tutorial Delivery: Conducted comprehensive tutorials explaining complex academic concepts, ensuring students developed a deeper understanding of subject matter. Assessment & Marking: Actively assisted in the preparation and marking of essays and assignments, providing constructive, detailed feedback to support student learning. Student Support: Provided extensive academic advisement to students regarding course selection, program pathways, and degree requirements. Curriculum Planning: Participated in faculty meetings regarding curriculum development, innovative teaching methodologies, and assessment strategies.	

AHEAD Operations	University of Sri Jayewardenepura, Sri Lanka
Project Assistant (ELTA-ELSE Faculty Grant)	Jan 2023 – Jun 2023
Grant Administration: Supported the coordination of the World Bank-funded AHEAD grant, ensuring alignment with academic project timelines and objectives. Event Management: Managed administrative logistics for academic workshops and research meetings.	

Department of Business Economics	University of Sri Jayewardenepura, Sri Lanka
Temporary Demonstrator	Sep 2022 – Nov 2022
Teaching: Facilitated interactive tutorial sessions for undergraduate students on economic concepts and analytical methodologies. Curriculum Design: Coordinated Outbound Training (OBT) programs and designed practical experiential learning activities aligned with course objectives. Examination Management: Provided invigilation services and assisted in organizing examination materials and schedules.	

Department of Business Economics	University of Sri Jayewardenepura, Sri Lanka
Temporary Research Assistant	Oct 2021 – Aug 2022
Methodology Training: Facilitated research methodology training sessions for junior researchers, explaining advanced econometric techniques and survey design. Statistical Tutoring: Conducted tutorial sessions on statistical software (Probit/Tobit models) for students and colleagues. Assessment: Evaluated research reports and analytical assignments, assessing the quality of econometric analysis and providing feedback.	

Centre for Forestry and Environment	University of Sri Jayewardenepura, Sri Lanka
Research Assistant	Apr 2018 – July 2021
Lecturing: Delivered guest lectures on Research Methodology to undergraduate honor students (2019-2021), covering questionnaire design, data collection, and statistics. Research Supervision: Directly supervised six undergraduate honors research projects, mentoring students through conceptualization, field data collection, and manuscript preparation. Assessment Design: Developed assessment materials including research proposals and methodology evaluations for advanced environmental science courses.	

Research Funding & Awards

ICGEB-DIC-MOST International Fellowship	2024 NSF SUSRED Award	2024
Postdoctoral Grant in ICGEB China RRC	Support Scheme for Supervision of Research Degrees	
Best Oral Presentation	2022 NRC Grant 18-032	2019
26th International Forestry and Environment Symposium	Research Assistant funding stream	

Indutrial Experience

Lion Brewery (Ceylon) PLC

Quality Assurance Executive Trainee

Biyagama, Sri Lanka

July 2018 – March 2019

- Quality Control: Investigated quality-related issues in apple-flavored beer production and recommended optimal pasteurization temperatures.
- Compliance: Maintained quality assurance standards throughout brewing and packaging processes, strictly adhering to ISO 9001 and ISO 22000 safety standards.
- Process Improvement: Participated in company-wide training on SAP-ERP systems and 5S workplace organization methodologies.

Education

Faculty of Graduate Studies

Pharmacognosy & Phytochemistry

Doctor of Philosophy (Ph.D.)

University of Sri Jayewardenepura, Sri Lanka • 2023

- Thesis:** Production of Mosquito Repellents Using Lesser Known, Under-Utilized Volatile Oil Producing Plant Species.
- Award:** NSF SUSRED Award 2024 (National Science Foundation award for high-impact doctoral research).

Faculty of Applied Sciences

Plant Biotechnology

Bachelor of Science (Hons) • Second Class

University of Sri Jayewardenepura, Sri Lanka • 2018

Thesis: *Evaluation of Microbial Counts in Apple Flavoured Beer Compared to Base Beer after Exposing to an External Environment.*

Publications

Patents

Munugoda, K. D., Subasinghe, S., Hapugoda, M. (2024). *Volatile Plant Oil-Based Mosquito Repellent Sticks & Sprays*. National Intellectual Property Office, Sri Lanka (App No: LK/P/1/23236).

Selected Journal Articles

- Munugoda, K. D., et al. (2023). Choice modelling for the commercial cultivation of underutilized aromatic plants for producing mosquito repellents. *Economic Botany*, 1-19.
- Chen, J., ..., Munugoda, K. D., et al. (2021). The complete chloroplast genome of Walla Patta, *Gyrinops walla* (Thymelaeaceae). *Mitochondrial DNA Part B*, 6(6), 1699-1701.

Manuscripts & Preprints

- Munugoda, K. D., et al. (2025). Reinventing Artisanal Curry Leaf Paste: Biochemical and Sensory Analysis. *agriRxiv* (Preprint).
- Munugoda, K. D., et al. (2024). Refining Nature's Defense: Production and Evaluation of Mosquito Repellents from *Ocimum* species. *ChemRxiv* (Preprint).

Memberships

Associate Member (A.I.Chem.C.)

Institute of Chemistry Ceylon

Membership No: 3548 Aff

English Language Proficiency

IELTS

Overall Band Score 7.0